Online Resource 1 to: *Shift-aware probabilities for random convolutional kernel time series classifiers via leverage-moderated leave-one-out calibration*

Online Resource 1 — Derivations

Notation. Training features $\varphi_i \in \mathbb{R}^p$ (MiniRocket output, standardised with the training mean and standard deviation), n training series, K classes, one-vs-rest targets $Y \in \{-1, +1\}^{n \times K}$. Augmented inputs $\tilde{x}_i = (1, \varphi_i - \bar{\varphi})$ with unpenalised intercept. Penalty matrix $P = \mathrm{diag}(0, \lambda I_p)$ and $A = \tilde{X}^T\tilde{X} + P$ ($\tilde{X}$ stacks the $\tilde{x}_i$). The ridge fit is $B = A^{-1}\tilde{X}^TY$, fitted logits $\hat{Y} = \tilde{X}B = HY$ with hat matrix $H = \tilde{X}A^{-1}\tilde{X}^T$, and $d_i = H_{ii}$.

Proposition 1 (exact LOO logits)

Let B_{-i} be the fit without example i, with the same λ and the same standardisation. Then

$$\tilde{x}_i^T B_{-i} = Y_i - (Y_i - \hat{Y}_i)/(1 - d_i).$$

Proof. $A_{-i} = A - \tilde{x}_i\tilde{x}_i^T$. By Sherman–Morrison, $A_{-i}^{-1} = A^{-1} + A^{-1}\tilde{x}_i\tilde{x}_i^TA^{-1}/(1 - d_i)$, so $\tilde{x}_i^TA_{-i}^{-1} = \tilde{x}_i^TA^{-1} + d_i\tilde{x}_i^TA^{-1}/(1 - d_i) = \tilde{x}_i^TA^{-1}/(1 - d_i)$. Since $B_{-i} = A_{-i}^{-1}(\tilde{X}^TY - \tilde{x}_iY_i)$, $\tilde{x}_i^TB_{-i} = (\tilde{x}_i^TA^{-1}\tilde{X}^TY - d_iY_i)/(1 - d_i) = (\hat{Y}_i - d_iY_i)/(1 - d_i) = Y_i - (Y_i - \hat{Y}_i)/(1 - d_i)$. ∎ (This is the classical PRESS identity of Allen (1974); PreVal uses the same identity.)

Proposition 2 (held-out leverage)

The leverage of training example i under the model trained without it is

$$h_{-i} := \tilde{x}_i^TA_{-i}^{-1}\tilde{x}_i = d_i/(1 - d_i).$$

Proof. From the Sherman–Morrison expression above, $\tilde{x}_i^TA_{-i}^{-1}\tilde{x}_i = d_i + d_i^2/(1 - d_i) = d_i/(1 - d_i)$. ∎

Proposition 3 (closed-form quantities from one thin SVD)

Let $\Phi c = USV^T$ be the thin SVD of the centred feature matrix (rank $r \le \min(n - 1, p)$), and $D = \mathrm{diag}(1/(s_j^2 + \lambda))$. For a test point with centred features ψ:

- $d_i = 1/n + \Sigma_j u_{ij}^2 s_j^2/(s_j^2 + \lambda)$;
- $h(x) := \tilde{x}^TA^{-1}\tilde{x} = 1/n + \|D\^{1/2}V^T\psi\|^2 + (\|\psi\|^2 - \|V^T\psi\|^2)/\lambda$;
- $k(x, i) := \tilde{x}^TA^{-1}\tilde{x}_i = 1/n + (V^T\psi)^T D S u_i$ (the training points lie in span(V)).

Proof. Because Φc is centred, the intercept column is orthogonal to Φc and A is block diagonal in the basis (1, Φc): the intercept block contributes 1/n. For the feature block, $(\Phi c^T\Phi c + \lambda I)^{-1} = V D V^T + \lambda^{-1}(I - VV^T)$. ∎

All three cost O(nr) per training point and O(pr) per test point after the SVD.

Proposition 4 (exact LOO model at a test point)

With residuals $E = Y - \hat{Y}$,

$$\tilde{x}(x)^TB_{-i} = \tilde{x}(x)^TB - k(x, i)\,E_i/(1 - d_i),\quad h_{-i}(x) = h(x) + k(x, i)^2/(1 - d_i).$$

Proof. $B_{-i} = A_{-i}^{-1}(\tilde{X}^TY - \tilde{x}_iY_i) = B - A^{-1}\tilde{x}_iY_i + A^{-1}\tilde{x}_i(\hat{Y}_i - d_iY_i)/(1 - d_i) = B - A^{-1}\tilde{x}_i(Y_i - \hat{Y}_i)/(1 - d_i) = B - A^{-1}\tilde{x}_iE_i/(1 - d_i)$; multiplying by $\tilde{x}(x)^T$ gives the first identity. The second is Sherman–Morrison applied to

$\tilde{x}(x)^T A_{-i}^{-1} \tilde{x}(x)$. ∎ (Both identities are checked numerically against explicit refits in `code/tests/test_core.py`.)

These give every quantity jackknife+ needs, $s(x, y; f_{-i})$ for all i, in O(nK) per test point, with no refitting. The jackknife+ set is $C(x) = \{y : \#\{i : s_i < s(x, y; f_{-i})\} < (1 - \alpha)(n + 1)\}$, where $s_i = s(x_i, y_i; f_{-i})$.

Interpretation of h(x)

Under the Bayesian linear model with prior $\beta \sim N(0, (\sigma^2/\lambda)I)$ and noise variance σ^2, the posterior predictive variance of each logit at x is $\sigma^2 \cdot h(x)$ (with the intercept treated as flat). The leverage-standardised logit $\tilde{x}(x)/\sqrt{h(x)}$ is therefore the posterior mean divided by its posterior standard deviation up to the common factor σ. In p − n the out-of-span term $(\|\psi\|^2 - \|V^T\psi\|^2)/\lambda$ dominates; exact held-out leverages h_{i} include the same term for each left-out example, so h(x)/h_ref ≈ 1 for in-distribution test inputs and grows for inputs away from the training distribution.

Caveats (stated in the paper)

- The identities are exact for fixed standardisation; the scaler and the MiniRocket biases are fitted on all n training series, so the LOO quantities ignore that dependence.
- λ and the temperature a are selected on the same LOO pairs, which makes the LOO NLL slightly optimistic and means the jackknife+ guarantee (≥ 1 − 2α, Barber et al. 2021) holds only approximately; coverage is therefore assessed empirically.